

Eric Lopez

Senior Data Engineer

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Summary

Senior Data Engineer with 11+ years of experience designing and scaling distributed data platforms supporting real-time analytics, enterprise reporting, and AI/ML workloads. Expertise in building **high-throughput streaming** and **batch data pipelines** using **Apache Kafka, Spark Streaming, Airflow, and dbt**, enabling reliable data processing at scale across **AWS and Azure cloud environments**. Specialized in **modern Lakehouse architectures (Delta Lake, Iceberg)** and **Snowflake-based analytics platforms**, implementing Medallion data models, automated ELT pipelines, and governed data layers to support enterprise data strategy. Proven ability to architect **fault-tolerant, event-driven data systems** that deliver low-latency processing, scalable ingestion, and high data reliability for mission-critical business workloads.

Experienced in **multi-cloud data platform modernization, infrastructure-as-code** using **Terraform**, and **distributed system optimization**, helping organizations transition legacy data environments into resilient, cloud-native analytics platforms that drive data-driven decision-making.

Skills

Programming & Data Processing: Python, SQL, PySpark, Scala, Bash, Shell Scripting

Streaming & Event-Driven Data Platforms: Apache Kafka, Spark Streaming, AWS Kinesis, RabbitMQ, Event-Driven Architecture

Data Engineering & Pipeline Orchestration: Apache Airflow, dbt, Apache NiFi, AWS Glue, Talend, ETL, ELT, Batch Processing, Real-Time Data Pipelines, Data Quality Monitoring

Big Data & Lakehouse Technologies: Databricks, Delta Lake, Apache Iceberg, Hadoop (HDP, HDF), Hive, Medallion Architecture, Distributed Data Processing

Data Warehousing & Analytics Platforms: Snowflake, Amazon Redshift, Azure Synapse Analytics, Amazon Athena

Databases & Data Stores: PostgreSQL, MySQL, Oracle, MongoDB, Amazon RDS, OpenSearch

Cloud Platforms: AWS (S3, EMR, Glue, Kinesis, Lambda, ECS, EC2, CloudFormation), Microsoft Azure (Data Factory, Data Lake Storage, Synapse Analytics, Blob Storage)

Infrastructure & DevOps: Docker, Terraform, CI/CD, GitHub Actions, Jenkins, Linux, Git

Data Governance & Security: Apache Ranger, Kerberos, LDAP, Row-Level Security, Column-Level Security, Data Governance

Business Intelligence & Analytics: Power BI, Tableau, Dashboard Development, KPI Reporting

Methodologies & Architecture: Agile, Scrum, SDLC, Dimensional Modeling, Data Architecture, Solution Architecture

Education

University of Washington | Seattle WA

Bachelor's degree in Data Engineering | September 2010 – May 2014

Relevant Coursework:

- Data Structures and Algorithms, Database Systems, Distributed Systems
- Data Warehousing & ETL Concepts, Big Data Systems, Cloud Computing Fundamentals
- Software Engineering Principles, Systems Design, Web Application Development
- Statistics and Introductory Machine Learning

Experience

Slalom | [Seattle, WA](#)

Senior Data Engineer | *October 2022 - Present*

Senior Data Engineer delivering end-to-end modern data platform solutions for U.S.-based enterprise clients. Partner with executive stakeholders to translate complex business requirements into scalable, low-latency, distributed data architectures aligned with enterprise data strategy and AI/ML readiness.

- Architected and implemented **Medallion Lakehouse architectures**, leveraging **dbt for transformation modeling and Airflow for pipeline orchestration**, enabling governed Bronze/Silver/Gold data layers across AWS and Azure.
- Designed and deployed real-time streaming pipelines using **Apache Kafka and Spark Streaming**, orchestrated through **Apache Airflow DAGs** to manage pipeline dependencies, retries, and SLA monitoring for high-throughput ingestion workflows.

- Engineered distributed, horizontally scalable data platforms optimized for performance, fault tolerance, and high availability across multi-cloud environments.
- Led modernization of enterprise data warehouses by migrating legacy reporting systems to **Snowflake**, implementing **dimensional data models and ELT pipelines** optimized for scalable analytics workloads.
- Designed **Snowflake ELT pipelines** using **Airflow** and **dbt**, enabling automated transformation, testing, and deployment of analytical data models.
- Implemented modular **dbt** transformation frameworks with incremental models, automated testing, and version-controlled analytics logic to standardize enterprise data transformations, reducing **ETL** pipeline failures by 40% and accelerating deployment cycles by 30% across 200+ tables.
- Developed semantic layers and lineage documentation in **dbt** to ensure data consistency and traceability across **Bronze/Silver/Gold** tables, resolving over 150 previously undetected data discrepancies and improving downstream reporting accuracy by 25%.
- Built automated data validation tests and monitoring within **dbt** to maintain high data quality, proactively detecting and alerting on 95% of data anomalies before production ingestion, which decreased manual error handling by 50 hours per month.
- Built scalable data pipelines supporting **feature engineering and ML-ready datasets**, enabling data science teams to train and deploy predictive models.
- Implemented event-driven architectures enabling low-latency data movement between **OLTP systems, streaming platforms, data lakes, and analytical warehouses**.
- Optimized Kafka streaming workloads through partition strategy redesign and parallel processing, increasing ingestion throughput by **3x and reducing latency to sub-second levels**.
- Designed and implemented a business-driven **Data Quality framework** with automated validation rule engines, monitoring, and alerting to improve enterprise data trust and governance compliance.
- Migrated legacy analytics and batch processing systems to modern cloud-native distributed architectures, improving performance, scalability, and operational reliability.
- Led development of modern BI solutions using Power BI, transforming static reports into interactive, automated, real-time dashboards driving executive decision-making.
- Defined enterprise data governance standards, aligning data architecture strategy with regulatory compliance and long-term business objectives.
- Collaborated with cross-functional teams (engineering, analytics, product, leadership) to design robust ETL/ELT and streaming data integration pipelines.
- Delivered knowledge transfer programs, architectural best practice frameworks, and technical mentorship to strengthen distributed systems and streaming capabilities across teams.

Skytap | [Seattle, WA](#)

Data Engineer | *January 2018 - September 2022*

As a Data Engineer at Skytap, led end-to-end delivery of data engineering projects, including the design and implementation of Lakehouse architectures on AWS and Azure. Built scalable data solutions leveraging Delta Lake and AWS Redshift to unify data processing and analytics. Worked extensively with core AWS technologies such as S3, Glue, Lambda, and EMR, gaining deep expertise in cloud-based data pipeline orchestration and performance optimization.

- Design and implementation of **Big Data Lakehouse architecture on AWS / Azure cloud**.
- Worked as Team Lead to implement complete **Lakehouse architecture including Orchestration and terraform CI/CD pipeline** and other serverless technologies.
- Implementation of reporting system
- Modern **data warehousing using Snowflake and Microsoft Azure Data Factory**.
- Migration of **Oracle database to data lake on Snowflake**
- Implemented **Data Validation checks using Rule Engine in Medallion Architecture**.
- AWS Redshift **security at row and column level**
- Real time streaming with **Kafka & Snowflake**
- Deployed **HDP/HDF Hadoop Cluster** & implemented **security & access control via Kerberos / LDAP / Ranger**
- Being flexible enough and adaptable to the changing and varied work settings.
- Designed architecture for Big Data pipelines to ingest data from disparate sources using **Spark / Delta Lake / Nifi and Kafka**.
- Designed & delivered ingestion pipelines and analytical workloads for various use cases using **Nifi/ Databricks / Delta Lake/Airflow / AWS and S3**.
- Designed and delivered **Big Data Analytic Frameworks using AWS EMR and Athena**.
- Developed real time streaming pipeline using **AWS Kinesis Data Streams, Kinesis Data firehose, S3, AWS Lambda and MongoDB**

- Worked in Agile model and participated in all agile ceremonies including Client Meetings, Scrum, Backlog Grooming, Sprint Planning, Retrospective, Release Planning Co-ordinating with QA
- Setup **10+ node Hadoop cluster** using **Hortonworks Data Platform (HDP)** and **Hortonworks Data Flow (HDF / NiFi / Kafka)**
- Development of Spark Streaming scripts for data analytics & anomaly detection.

Deloitte | [Seattle, WA](#)

Data Engineer | *September 2014 - December 2017*

After graduating from **University of Washington** with a degree in **Computer Science**, I began my career as a **Data Engineer** in a high-growth environment.

- Over three years, I built a solid foundation in designing and maintaining data platforms that powered data-driven financial operations and analytics.
- Contributed to the development of **online invoice factoring** and **working capital financing** systems, enabling small businesses to make faster, data-informed funding decisions.
- Designed and managed **data infrastructure** components: databases, data lakes, and ETL pipelines—to ensure secure, reliable, and scalable management of financial datasets.
- Developed **ETL pipelines** using **Python** and **Apache Spark** for cleansing, transforming, and standardizing raw data into analytics- and ML-ready formats.
- Integrated and centralized **financial and transactional data** into **Amazon Redshift**, enabling near real-time reporting and accurate KPI tracking for stakeholders.
- Improved **pipeline and query performance**, reducing processing time and enhancing system efficiency across large-scale datasets.
- Collaborated cross-functionally with **analytics, product, and finance teams** to translate data into actionable insights while upholding strict **data governance, security, and compliance** standards.